Healthcare Claims and Fraud Analysis

Exploratory Data Analysis

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Source: https://www.kaggle.com/datasets/bonifacechosen/nhis-healthcare-claims-and-fraud-dataset/versions/1

Dataset Overview

1,500
Total Claims

9
Features Tracked

22
Unique Diagnoses

0
Missing / Duplicate Rows

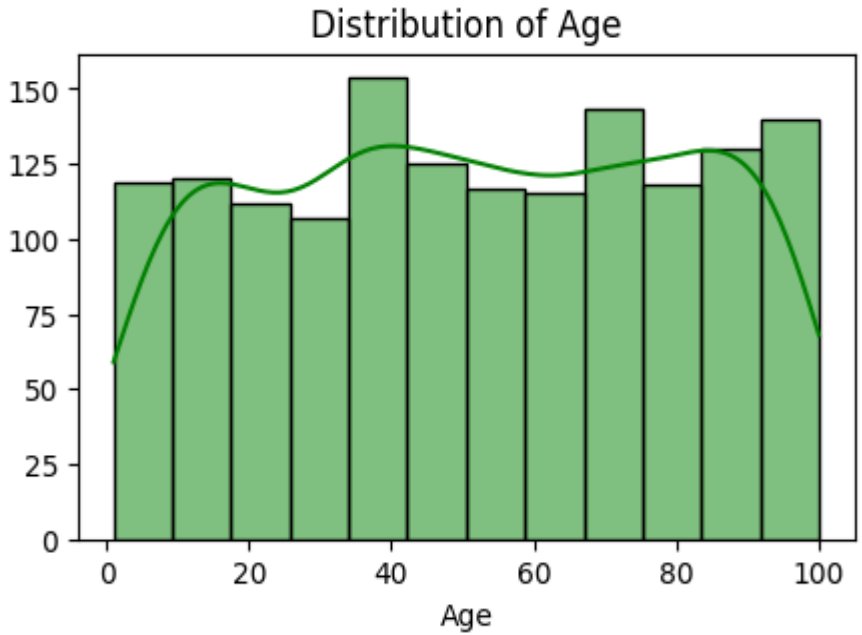
Columns Analysed

- Patient ID · Age · Gender
- Date Admitted · Date Discharged
- Diagnosis · Treatment
- Amount Billed · Fraud Type
- Derived: Length of Stay · Cost per Day

Data Quality Summary

✓ No missing values across all 9 columns
✓ No duplicate records detected
✓ All data types validated correctly
✓ Dates converted to datetime format
✓ Derived features engineered successfully

Patient Demographics & Billing Overview



₹3,17,657

Avg Amount Billed

4.8 days

Avg Length of Stay

₹84,990

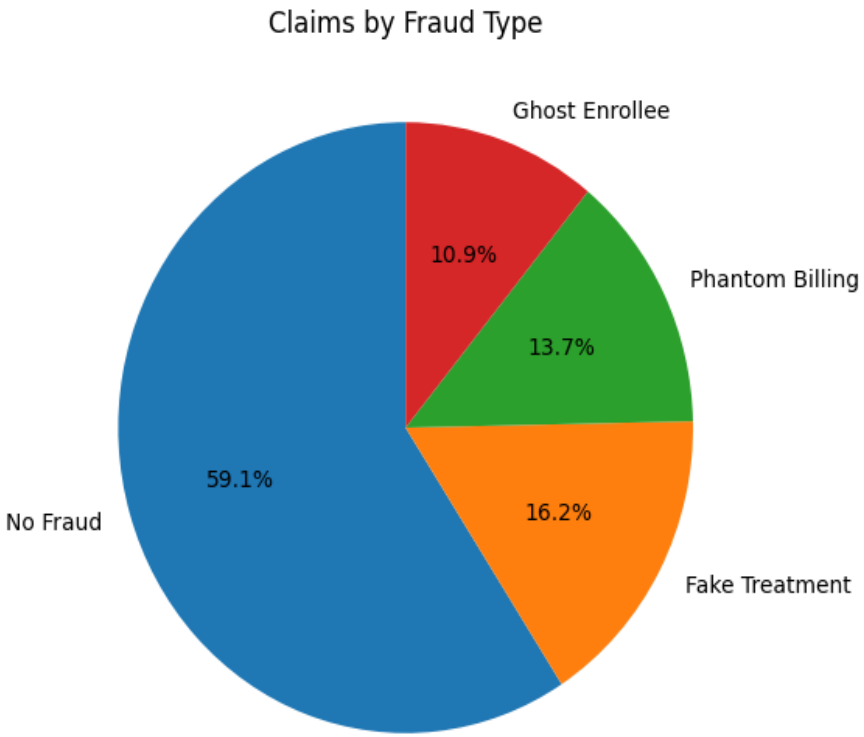
Avg Cost per Day

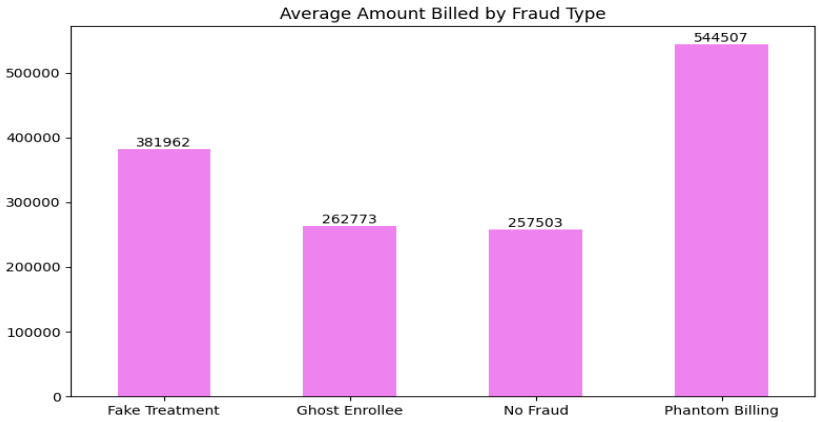
51 yrs

Median Patient Age

Gender Split:
Female avg billing: ₹3,24,757
Male avg billing: ₹3,10,725

Fraud Type Analysis





Insights:

- Phantom Billing has by far the highest average billed amount at 544,507 which is more than 2.1× the average legitimate claim (257,503).
- Fake Treatment averages 381,962, about 48% higher than legitimate claims.
- Ghost Enrollee (262,773) is closest to legitimate claim values, making it harder to detect through billing amount alone.
- Phantom Billing is clearly the most financially damaging fraud type per claim.

High-Cost Diagnoses & Fraud Hotspots

Top 10 High Cost Diagnosis:

Diagnosis	Amount Billed
Epilepsy Surgery	625052.184783
Infertility Treatment (IVF)	583118.363030
Cosmetic Surgery	581171.548148
Organ Transplant	579659.167826
Complex Heart Surgery	506539.412692
Neurosurgery	503446.875238
Cancer Treatment	491025.682333
Advanced Spinal Surgery	480536.627391
Cesarean Section	305776.758333
Gastroenteritis	300760.917889

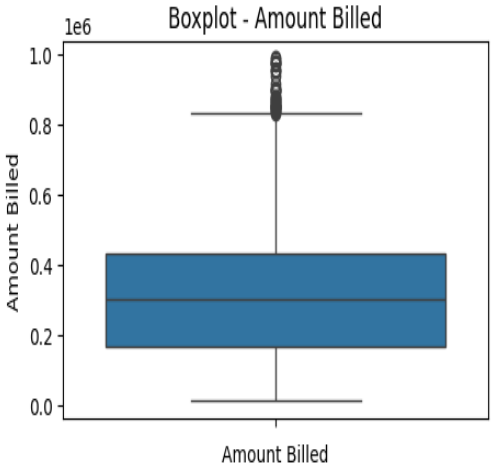
Top 10 Flagged Diagnosis:

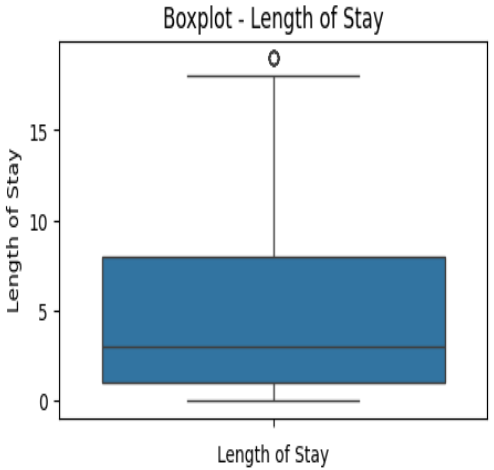
Diagnosis	
Tuberculosis	40
Hypertension	35
Pneumonia	34
Infertility Treatment (IVF)	33
Gastroenteritis	33
HIV/AIDS	32
Cancer Treatment	30
Cesarean Section	29
Stroke	28
Peptic Ulcer	28

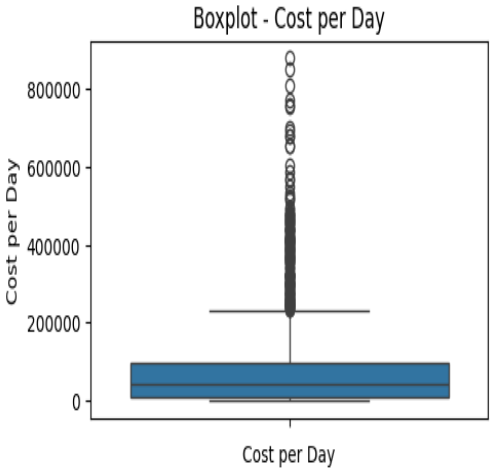
Insights:

- Epilepsy Surgery (625,052), IVF (583,118), and Cosmetic Surgery (581,172) are the three most expensive diagnoses on average.
- Tuberculosis (40 cases), Hypertension (35), and Pneumonia (34) have the highest absolute fraud flag counts.
- These are common, high-volume conditions and their fraud frequency likely reflects the ease of fabricating routine diagnoses.
- Interestingly, common chronic conditions (Hypertension, Diabetes, Peptic Ulcer) appear alongside infectious diseases, suggesting fraudsters exploit high-prevalence diagnoses to blend in.

Outlier Detection





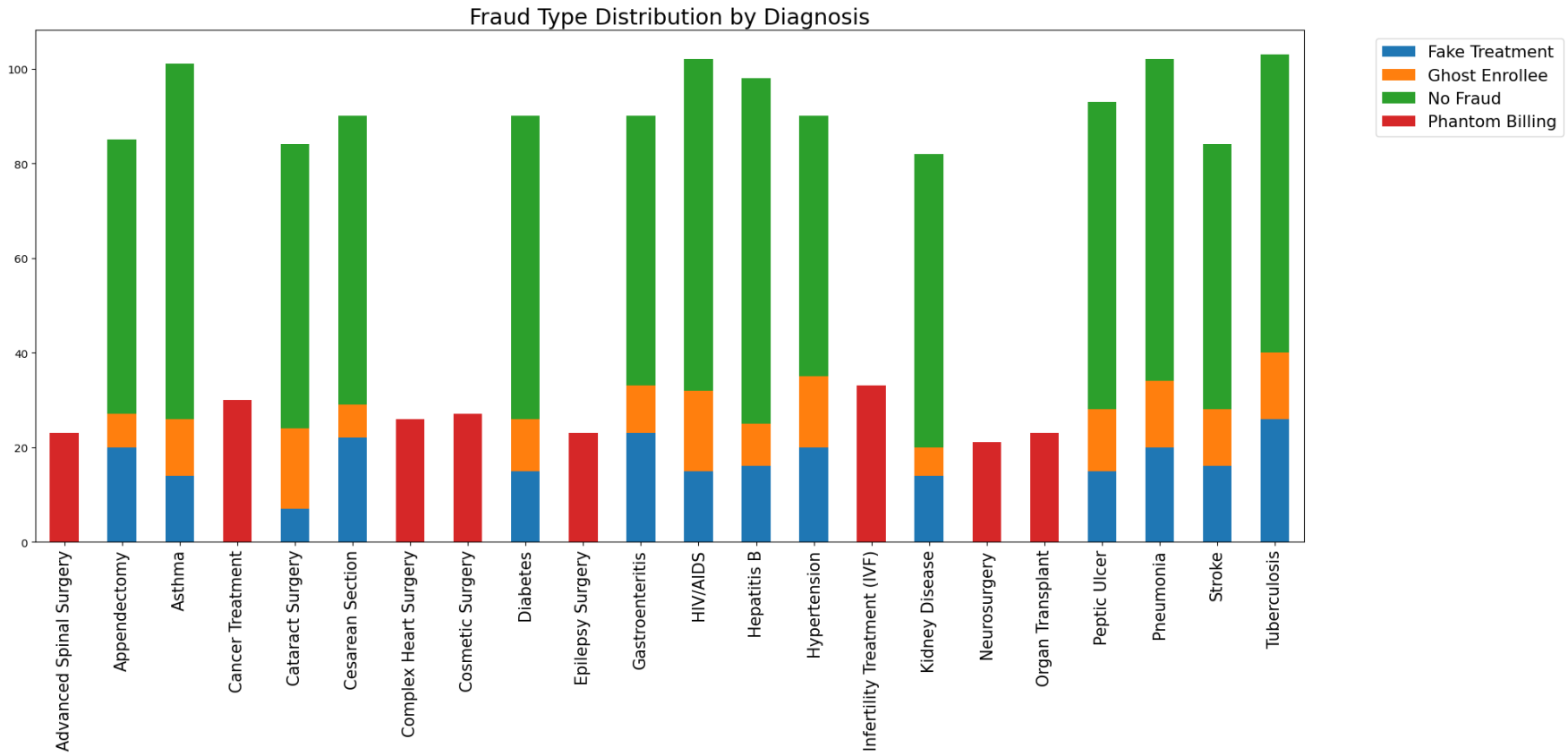


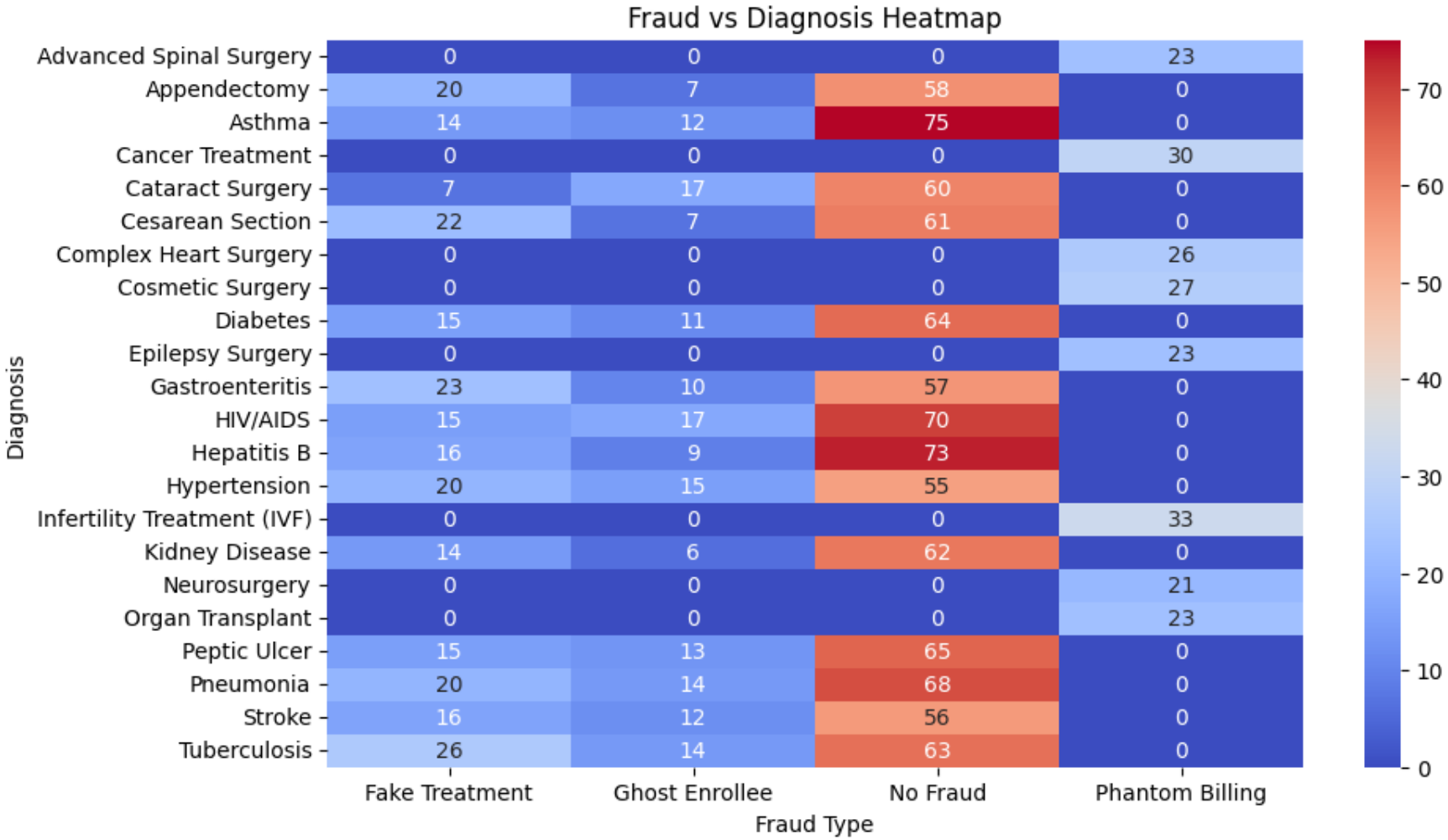
Insight:

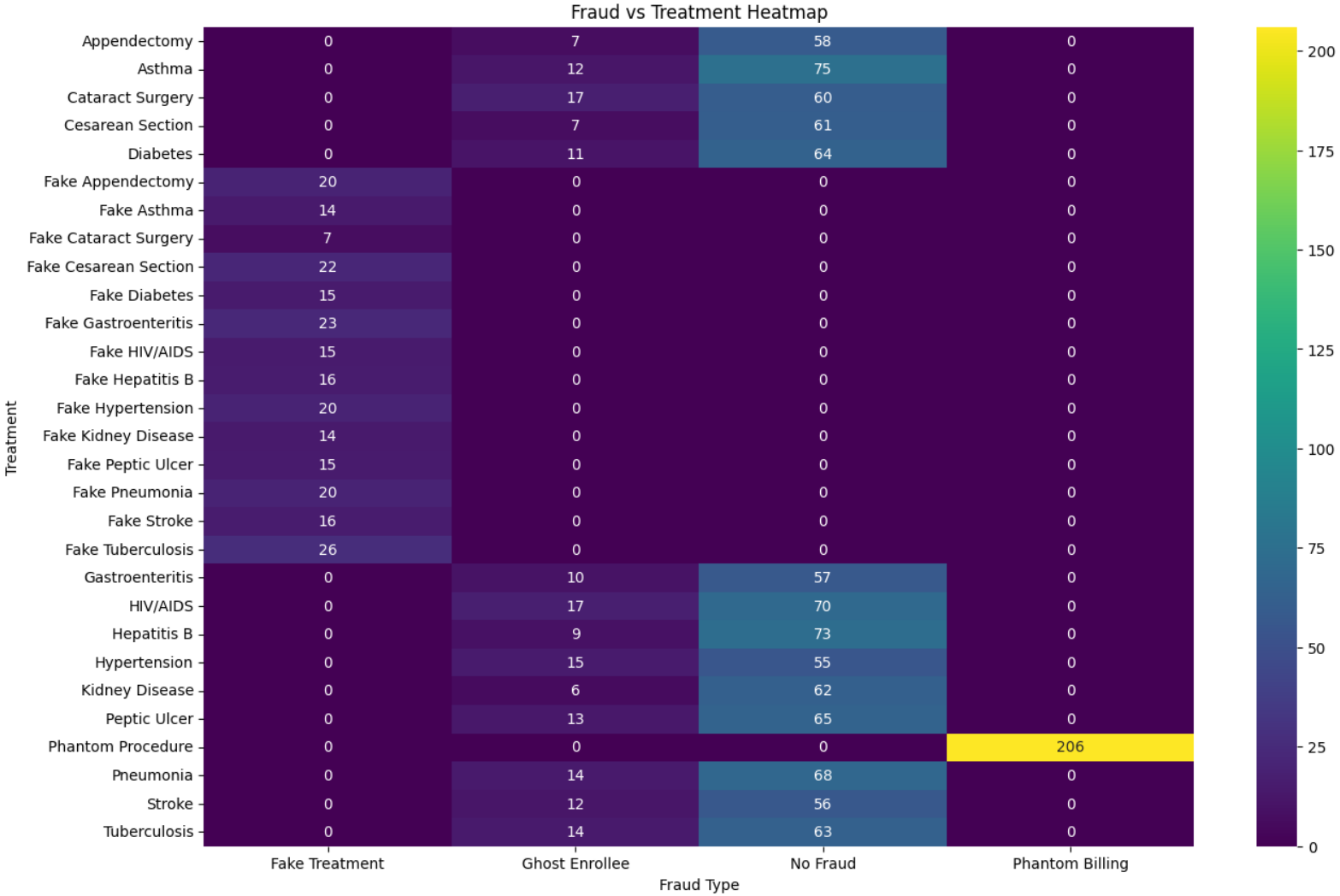
A small subset of claims shows disproportionately high costs, making them prime candidates for **fraud investigation and audit prioritization**.

Fraud Mechanism Analysis

Fraud Type	No. of Claims	Avg Amount (₹)	Key Insight
Phantom Billing	206	₹5,44,507	All linked to a single "Phantom Procedure" treatment code. It is the most financially damaging fraud type.
Fake Treatment	237	₹3,81,962	19 fake treatment variants identified. Fake Tuberculosis (26) and Fake Cesarean Section (22) most prevalent.
Ghost Enrollee	124	₹2,62,773	Patients enrolled without evidence of care. Lowest avg cost.


Fraud Type Distribution by Diagnosis
Fake Treatment
Ghost Enrollee
No Fraud
Phantom Billing
0
20
40
60
80
100
Advanced Spinal Surgery
Appendectomy
Asthma
Cancer Treatment
Cataract Surgery
Cesarean Section
Complex Heart Surgery
Cosmetic Surgery
Diabetes
Epilepsy Surgery
Gastroenteritis
HIV/AIDS
Hepatitis B
Hypertension
Infertility Treatment (IVF)
Kidney Disease
Neurosurgery
Organ Transplant
Peptic Ulcer
Pneumonia
Stroke
Tuberculosis


Fraud vs Diagnosis Heatmap
Diagnosis
Advanced Spinal Surgery 0 0 0 23
Appendectomy 20 7 58 0
Asthma 14 12 75 0
Cancer Treatment 0 0 0 30
Cataract Surgery 7 17 60 0
Cesarean Section 22 7 61 0
Complex Heart Surgery 0 0 0 26
Cosmetic Surgery 0 0 0 27
Diabetes 15 11 64 0
Epilepsy Surgery 0 0 0 23
Gastroenteritis 23 10 57 0
HIV/AIDS 15 17 70 0
Hepatitis B 16 9 73 0
Hypertension 20 15 55 0
Infertility Treatment (IVF) 0 0 0 33
Kidney Disease 14 6 62 0
Neurosurgery 0 0 0 21
Organ Transplant 0 0 0 23
Peptic Ulcer 15 13 65 0
Pneumonia 20 14 68 0
Stroke 16 12 56 0
Tuberculosis 26 14 63 0
Fake Treatment
Ghost Enrollee
No Fraud
Phantom Billing
Fraud Type
70
60
50
40
30
20
10
0


Fraud vs Treatment Heatmap
Treatment
Fraud Type
Fake Treatment
Ghost Enrollee
No Fraud
Phantom Billing
Appendectomy 0 7 58 0
Asthma 0 12 75 0
Cataract Surgery 0 17 60 0
Cesarean Section 0 7 61 0
Diabetes 0 11 64 0
Fake Appendectomy 20 0 0 0
Fake Asthma 14 0 0 0
Fake Cataract Surgery 7 0 0 0
Fake Cesarean Section 22 0 0 0
Fake Diabetes 15 0 0 0
Fake Gastroenteritis 23 0 0 0
Fake HIV/AIDS 15 0 0 0
Fake Hepatitis B 16 0 0 0
Fake Hypertension 20 0 0 0
Fake Kidney Disease 14 0 0 0
Fake Peptic Ulcer 15 0 0 0
Fake Pneumonia 20 0 0 0
Fake Stroke 16 0 0 0
Fake Tuberculosis 26 0 0 0
Gastroenteritis 0 10 57 0
HIV/AIDS 0 17 70 0
Hepatitis B 0 9 73 0
Hypertension 0 15 55 0
Kidney Disease 0 6 62 0
Peptic Ulcer 0 13 65 0
Phantom Procedure 0 0 0 206
Pneumonia 0 14 68 0
Stroke 0 12 56 0
Tuberculosis 0 14 63 0
200
175
150
125
100
75
50
25
0

Operational Metrics: Length of Stay & Cost per Day

Avg Length of Stay by Fraud Type

Fraud Type	Length of Stay
Fake Treatment	5.020576
Ghost Enrollee	4.317073
No Fraud	4.828636
Phantom Billing	4.941748

Avg Cost per Day by Fraud Type

Fraud Type	Cost per Day
Fake Treatment	94440.065763
Ghost Enrollee	73951.057868
No Fraud	69851.514244
Phantom Billing	147816.160783

Insight:

Fake Treatment carries the highest Cost per Day (about ₹95K) as well as Length of Stay, suggesting inflated daily charges as the primary billing manipulation lever.

Key Findings & Recommendations

The exploratory data analysis reveals clear patterns that can help in identifying fraudulent healthcare claims.

- The dataset shows **significant right-skewness in financial variables**, especially *Amount Billed* and *Cost per Day*, indicating the presence of unusually high-cost claims.
- **Amount Billed** is mostly within normal range, but several high-value outliers suggest potential overbilling.
- **Cost per Day emerges as the strongest fraud indicator**, with extreme outliers suggesting possible overbilling or manipulation.
- **Phantom Billing poses maximum financial risk.** 206 claims entirely under one code with an average of ₹5.4L which is 2.1× above legitimate claims. Immediate audit required.
- A small subset of claims shows disproportionately high costs, making them prime candidates for **fraud investigation and audit prioritization**.
- The absence of missing data improves reliability, but the presence of **extreme variability** highlights the need for robust fraud detection mechanisms.